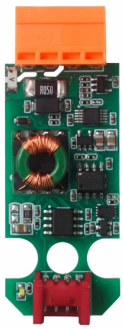
Unit AMeter

SKU:U086













Description

Unit AMeter is a current meter that can monitor current in real-time. It uses a 16-bit ADC digital-to-analog converter ADS1115 internally and communicates via I2C (0X48). To ensure measurement accuracy, it has a built-in DC-DC isolated power supply, and the I2C interface is electrically isolated using a low-power isolator CA-IS3020S to prevent noise and surges on the data bus or other circuits from entering the local ground and interfering with or damaging sensitive circuits. Each unit is individually calibrated at the factory, with an accuracy of 1% of full scale, ±1 digit reading, a resolution of 0.3mA, and a maximum measurement current of ±4A. It has an internal 4A fuse to prevent circuit damage from excessive measurement current.

Note

Calibration Parameters

EEPROM (0x51) has calibration parameters built-in at the factory. Please do not write to the EEPROM, as this will overwrite the calibration data and result in inaccurate measurements.

Features

- ±4A range

- 16-bit ADC conversion
- 4A fuse
- Accuracy of 1% of full scale, ±1 digit reading
- Resolution of 0.3mA
- LED power indicator
- Overcurrent protection
- No recalibration required (EEPROM factory written)
- Built-in CA-IS3020S isolation chip, anti-interference
- Isolated DC-DC
- Development platforms: Arduino, UIFlow (Blockly, Python)
- 2 x LEGO compatible holes

Includes

- 1 x Unit AMeter
- 1 x HY2.0-4P Grove cable (20cm)
- 1 x HT3.96-4P Terminal

Applications

- Ammeter

Specifications

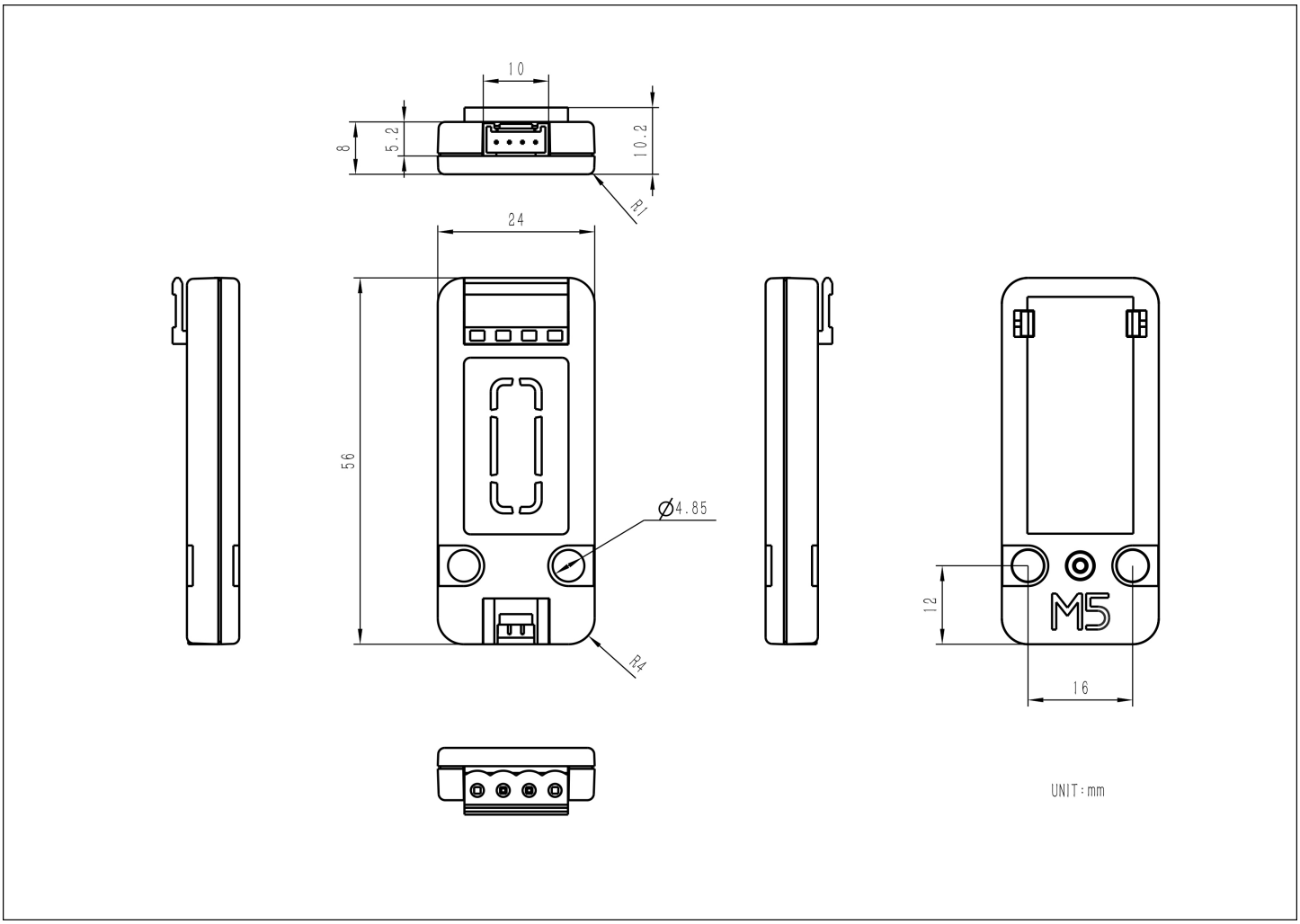
Specification	Parameter
Measurement Range	±4A
Communication Interface	I2C @0x48
Product Size	56.0 x 24.0 x 8.0mm
Product Weight	8.4g
Package Size	138.0 x 93.0 x 9.0mm
Gross Weight	16.7g

PinMap

Unit AMeter

HY2.0-4P	Black	Red	Yellow	White
PORT.A	GND	5V	SDA	SCL

Model Size



Structure

- Unit AMeter Structure Files

Datasheets

- CA-IS3020S
- ADS1115

Softwares

Arduino

- M5Unit-Ameter with M5Core
- M5Unit-Ameter with M5Core2
- M5Unit-Ameter with M5Atom
- M5Unit-Ameter with M5StickC
- M5Unit-Ameter with M5StickCPlus

UiFlow1

- Unit AMeter UiFlow1 Docs
- Unit AMeter UiFlow1 Example

UiFlow2

- coming soon...

Home Assistant

- Unit AMeter Home Assistant Integration

EasyLoader

Easyloader	Download Link	Note
Unit AMeter example with M5Core	download	/

Video

AMeter.mp4